24V DC POWER IN
(Main Supply from U1)

• Pin 1: +24V DC
• Pin 2: Power GND

+24V

GND In

MP1584 BUCK CONVERTER
24V In ➔ 5.00V DC Out

• IN+ | IN- (GND)
• OUT+ (5V) | OUT- (GND)

+24V +24V (DC+)

+5.0V (VIN)

GND (OUT-)

COMMON GROUND BUS (GND)
[Star-ground distribution]

GND (Black)

GND

IN+

GND

MP1584 BUCK CONVERTER
24V In ➔ 5.00V DC Out

• IN+ | IN- (GND)
• OUT+ (5V) | OUT- (GND)

+5.0V (Red)

GND

MG90S METAL SERVO
(Exhaust Butterfly Flap)

• Red: +5.0V (from Buck)
• Brown: GND (Common)
• Yellow: Signal (GPIO 18)

PWM (GPIO 18)

NOCTUA NF-F12 24V PWM
(120mm Exhaust Fan)

• Pin 1 (Yellow): +24V DC
• Pin 2 (Black): GND
• Pin 3 (Green): Tach / RPM
• Pin 4 (Blue): PWM (GPIO 19)

SSD1306 0.96" OLED
(I2C: 0x3C • Yellow / Blue)

• Pin 1: GND
• Pin 2: VCC (3.3V)
• Pin 3: SCL (GPIO 22)
• Pin 4: SDA (GPIO 21)

+3.3V (VCC)

GND

SDA (GPIO 21)

SCL (GPIO 22)

ESP-WROOM-32 (30-PIN)
On Terminal Shield

[LEFT TERMINALS]
• VIN (5.0V Power In)
• GND (System Ground)
• 3V3 (3.3V Out)
• GPIO 21 (I2C SDA)
• GPIO 22 (I2C SCL)
• GPIO 34 (ADC: 5-Way Switch)

[RIGHT TERMINALS]
• GPIO 18 (PWM: Flap Servo)
• GPIO 19 (PWM: Noctua 24V)
• GPIO 23 (Out: Blower MOSFET)
• GPIO 25 (Out: 110V Heater Relay)

[WiFi: Moonraker REST API]

PWM (GPIO 19)

GND (DC-)

Gate (GPIO 23)

NOYITO AOD4184 MOSFET
(Optocoupler Isolated)

• DC+: +24V | DC-: GND
• IN+: GPIO 23 | IN-: GND
• OUT+: +24V Pass-thru
• OUT-: Switched Motor (-)

+24V (OUT+)

Switched GND (OUT-)

24V BLOWER FAN
(Carbon Recirculation)

• Red Wire (+): ➔ OUT+
• Black Wire (-): ➔ OUT-

RS1010 5-WAY SWITCH
(1-Pin Resistor Ladder)

• Top Supply: 3.3V
• Resistors: 4x 10kΩ in series
• Pull-down: 100kΩ to GND
• Pin 1 (Wiper) ➔ GPIO 34

[1:OFF | 2:LOW | 3:MED | 4:HI | 5:PURGE]

GND (Ladder)

+3.3V (Ladder Top)

Analog (GPIO 34)

Heater (GPIO 25)

⚠ 110V AC MAINS HIGH-VOLTAGE ISOLATED ZONE
(Physically separated from all low-voltage lines)

AEDIKO 24V RELAY MODULE
(Optocoupler Isolated • 10A 250VAC)

• DC+: +24V | DC-: GND
• IN: shorted to +24V (jumper - H)
• COM: 110V Live In
• NO: 110V Switched Live Out

IN

DC+

DC-

AC

Switched Live (NO)

G3MB-202P 5V SSR MODULE
- DC+ 5v
- DC- GND
- IN GPIO 25
- AC - Load

IN

DC-

DC+

GND (DC-)

Series Live

NEON PILOT LIGHT
(Wired in parallel across PTC)

KSD301 70°C THERMOSTAT
[MANUAL RESET, NC]
Chamber Air Safety Cutoff

AC

121°C THERMAL FUSE (SF119E)
[ONE-SHOT TCO CUTOFF]
Heater Intake Meltdown Safety

Fused Live Out

PTC HEATER ELEMENT
110V AC, 100W–150W

• Terminal L: Fused Live
• Terminal N: Neutral

110V Live (Black)

110V AC MAINS INFEED
(Wall Cord)

• Live (L): Black Wire
• Neutral (N): White Wire
• Earth (PE): Green Wire

110V Neutral (White)

Earth Ground (Green)

8-pin connector - out
- + 24v
- Servo PWM
- Exhaust fan PWM
- GND
- + 3.3v
- Display SCL
- Discpal SDA
- Mode selecctor

8-pin connector - power
- AC N
- AC L
- KSD301
- KSD301
-
-
- +24v in
- GND